# Rivan Wong 黃家輝

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## Summary

M.S. in Computer Science who builds software and the systems that verify it. Two and a half years of internship and industry co-op experience at Taiwan Mobile and Sunbird Software spanning automated testing, API testing, and technical project management; thesis research on federated learning security, with six competing defenses reimplemented in PyTorch.

## Education

**National Taipei University of Technology (國立臺北科技大學)** *Taipei, Taiwan*
*M.S. in Computer Science and Information Engineering* *2025 – 2026*
- GPA 4.0/4.0. Completed in one year through the university's selective Bachelor's–Master's program (學碩一貫學程); thesis advised by Shiang-Jiun Chen, Ph.D., in the Software Security Lab.

**National Taipei University of Technology (國立臺北科技大學)** *Taipei, Taiwan*
*B.S. in Computer Science and Information Engineering* *2021 – 2025*
- GPA 3.5/4.0. Member of the Software Security Lab, focused on test automation.

## Languages

**Indonesian:** Native | **English:** Fluent (TOEIC 965) | **Chinese (Mandarin):** Working Proficiency (TOCFL B2), both degrees in Chinese

## Technical Skills

**Programming Languages**: Python · C++ · Java · HTML · CSS
**Testing & QA**: Robot Framework · Selenium · Appium · PyTest · Parallel-safe test design · Test case design · UAT
**Tools & Platforms**: Git · GitHub · Jenkins · Jira · Notion · VS Code · REST APIs · Linux (Ubuntu) · Windows
**Machine Learning & Research**: PyTorch · Flower · Federated learning · Adversarial ML
**Methodologies**: Agile (Scrum/Kanban) · CI/CD · Acceptance-test-driven development · OOP · SDLC

## Experience

**Taiwan Mobile Co., Ltd. (台灣大哥大股份有限公司)** *Taipei, Taiwan*
*Technical Project Management & QA Intern* *Jul 2025 – Jun 2026 · 12 months*
- Built a PyTest suite from scratch over an internal service's critical endpoints, working only from its API documentation; also fixed a colleague's Java API behind an existing Jenkins job, and built a pipeline, adapted from an existing one, that runs the checks and files results in the team's tracker automatically.
- Owned pre-release verification for two internal AI products (a VS Code extension that reviews code against company development standards, and a browser extension that translates content in an internal authoring portal), catching defects that compiled cleanly and looked right in the diff: dropped hyperlinks, truncated model output. Reported each to its engineer and wrote the test reports.
- Owned the release gate for the team building them: proposal and design agreed before code, explicit written acceptance criteria, then hands-on sign-off, across roughly 40 specified changes and 300+ criteria in about two months.
- Coordinated the six-person team that built both products across two countries, as the technical link between the on-site engineers and the remote interns, managing sprint tasks in Jira so both sides worked from one view of state.

**Sunbird Software Inc., Taiwan Branch (美商太陽鳥軟體股份有限公司台灣分公司)** *New Taipei, Taiwan*
*Test Automation Intern, Industry-Academia Co-op via the Software Security Lab* *Jul 2023 – Dec 2024 · 18 months*
- Turned ticket-level requirement specifications into Selenium-driven end-to-end suites for an enterprise data-center web platform whose 730+ suites run as an overnight regression, composed from a shared library of thousands of reusable Robot Framework keywords, with state set up through a REST API instead of slow UI flows.
- Kept hundreds of suites parallel-safe on one live server by declaring, by tag, the shared subsystems each touches.
- Cleared the same merge gate every time: zero warnings from custom lint rules, ten consecutive green runs, then five more under deliberate network throttling. Spent summer 2024 embedded full-time in the company's office.

## Research & Projects

**L-MAD: Trust-Aware Layer-Wise Median Absolute Deviation** Master's Thesis · 2025 – 2026
- Designed a two-stage defense for Byzantine-robust aggregation in federated edge learning that rejects poisoned layers individually instead of discarding the whole client, holding 97.5% DDoS-class $F_1$ under a 30% poisoning attack: 25.7 points clear of the strongest baseline and 2.6× faster than the nearest layer-wise defense.
- Implemented the method and reimplemented all six competing defenses in one harness in PyTorch and Flower, then ran 21 experiment groups over 20 seeds with significance testing, closing with the five conditions under which the method fails.

**Test Automation Best Practices: A Case Study Analysis** Capstone Project · 2023 – 2024
- With two teammates inside the Sunbird co-op, documented the seven gates a commercial test automation team takes a ticket through to a merged pull request, then analyzed four years of its output: down about two thirds from its peak, process unchanged.

**Sky Scraper Escape** OOP Lab Course · C++, MFC · Mar – Jun 2024
- Fit seven levels and the menus onto a teaching framework that allows exactly three game states, by mirroring its state interface onto a base class so every level and menu became an interchangeable object behind one pointer.
- Wrote 76% of the 2,237 lines of C++ the two of us authored (base class, player, menu system), merged through 17 pull requests.

Chinese version on the reverse side.

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## 摘要

資訊工程碩士，打造軟體，也打造驗證軟體的測試系統。曾在台灣大哥大與 Sunbird Software 累積兩年半的實習與產學合作經驗，涵蓋自動化測試、API 測試與技術專案管理；碩士論文研究聯邦式學習安全，並以 PyTorch 重新實作六套競爭防禦方法。

## 學歷

**國立臺北科技大學** 臺灣臺北
資訊工程碩士 *2025 – 2026*
- GPA：4.0 / 4.0。透過甄選制學碩一貫學程，以一年完成碩士學位；碩士論文於軟體安全實驗室完成，指導教授為陳香君博士。

**國立臺北科技大學** 臺灣臺北
資訊工程學士 *2021 – 2025*
- GPA：3.5 / 4.0。軟體安全實驗室成員，專注於自動化測試。

## 語言能力

**印尼文：**母語 | **英文：**流利（TOEIC 965）| **中文：**高階（TOCFL B2），學士與碩士皆以中文完成

## 專業技能

**程式語言**：Python · C++ · Java · HTML · CSS
**測試與品質保證**：Robot Framework · Selenium · Appium · PyTest · 可安全平行執行的測試 · 測試案例設計 · UAT
**工具與平台**：Git · GitHub · Jenkins · Jira · Notion · VS Code · REST API · Linux（Ubuntu） · Windows
**機器學習與研究**：PyTorch · Flower · 聯邦式學習 · 對抗式機器學習
**方法論與概念**：敏捷開發（Scrum/Kanban） · CI/CD · 驗收測試驅動開發 · OOP · SDLC

## 經歷

**台灣大哥大股份有限公司** 臺灣臺北
技術專案管理與 *QA* 實習生 *2025* 年 *7* 月 – *2026* 年 *6* 月 · *12* 個月
- 從零建置 PyTest 測試套件，涵蓋某項內部服務的關鍵 API 端點，僅依據 API 文件寫成；也修正同事既有 Jenkins 作業背後的 Java API，並仿照既有作業建置自動執行檢查、將結果建立成工單的 pipeline。
- 負責兩項內部 AI 產品的上線前驗證：依公司開發規範審查程式碼的 VS Code 擴充套件，以及在內部內容編寫平台旁完成翻譯工作的瀏覽器擴充套件；找出能順利編譯、在 diff 中看起來也正確的缺陷，包括譯文中遺失的超連結與中途截斷的模型輸出，並回報給開發工程師、撰寫測試報告。
- 掌管團隊的驗證關卡：先取得提案與設計共識再動手寫程式，訂出明確的書面驗收標準，最後在發布前親自驗證並簽核；兩個月左右涵蓋約 40 項規格化變更與 300 條以上的驗收條件。
- 擔任現場工程師與遠端實習生的技術橋樑，協調分處兩國的六人團隊，並以 Jira 管理 sprint 任務，讓雙方依據同一份狀態工作。

**美商太陽鳥軟體股份有限公司台灣分公司** 臺灣新北
自動化測試實習生（經由軟體安全實驗室的產學合作） *2023* 年 *7* 月 – *2024* 年 *12* 月 · *18* 個月
- 把工單層級的需求規格轉成由 Selenium 驅動的端對端測試套件，對象為企業級資料中心 Web 平台，其 730 組以上套件每晚平行執行回歸測試；套件以共用函式庫中數千個可重複使用的 Robot Framework 關鍵字組合而成，並透過 REST API 建立測試狀態，取代緩慢的使用者介面操作流程。
- 撰寫可安全平行執行的測試：以標籤宣告每個套件會動到的共用子系統，讓數百套件同時跑在同一台線上伺服器不互相干擾。
- 每一次合併都通過同一道關卡：自訂 lint 規則零警告、連續十次執行全數通過，接著再刻意限制網路頻寬多跑五次。2024 年夏季以全職實習生身分進駐公司辦公室。

## 研究與專案

**基於信任感知之逐層中位數絕對偏差（L-MAD）** 碩士論文 · 2025 – 2026
- 設計 L-MAD，一套用於聯邦邊緣學習的兩階段拜占庭強健聚合機制；逐層拒斥遭毒化的層，而不是丟掉整個用戶端。在 30% 毒化攻擊下，DDoS 類別 $F_1$ 維持在 97.5%，領先最強基準 25.7 個百分點，速度為最接近的逐層防禦的 2.6×。
- 以 PyTorch 與 Flower 實作方法，並在同一套實驗框架中重新實作六套競爭防禦方法；執行 21 組實驗、20 組隨機種子與顯著性檢定，最後整理出方法失效的五項條件。

**Test Automation Best Practices: A Case Study Analysis** 畢業專題 · 2023 – 2024
- 和兩位隊友在 Sunbird 產學合作第一線，逐道記錄商業自動化測試團隊從需求工單到合併 pull request 的七道關卡；再分析團隊四年的測試案例產出，發現流程未變，產出卻比高峰期下降約三分之二。

**Sky Scraper Escape** 物件導向實習課 · C++、MFC · 2024 年 3 – 6 月
- 在只允許三種遊戲狀態的教學框架中完成七個關卡與選單；將框架的狀態介面對應到自行設計的關卡基底類別，讓每個關卡與選單都成為同一個指標背後可互換的物件。
- 兩人共寫 2,237 行 C++，其中 76% 由我完成（關卡基底類別、玩家物件、選單系統），經 17 個 pull request 整合。

英文版請見背面。